https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	A is the independent variable and *f* is the dependent variable **or** vary *A* and measure *f*	**1**
	keep *M* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • elastic cord fixed at one end to a support • other end passed over a pulley • labelled pulley • labelled load	**1**
	vibrator connected to signal generator	**1**
	increase/decrease the frequency of the signal generator until stationary wave pattern is observed	**1**
	measure diameter of cord with micrometer/calipers	**1**
	Method of analysis	
	plot a graph of f^2 against 1 / *A* or equivalent (e.g. lg *f* against lg *A*)	**1**
	relationship valid if a straight line passing through the origin is produced (for lg *f* against lg *A*, relationship valid if a straight line with gradient –½)	**1**
	$k = \dfrac{M}{\text{gradient} \times 4 \times L^2}$ (for lg *f* against lg *A*, $k = M / [10^{(2 \times y\text{-intercept})} \times 4 \times L^2]$)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 use safety goggles/safety screen <u>to prevent injury to eyes from (moving) elastic cord/load</u> **or** use cushion/sand box <u>in case load falls</u>	
	D2 keep *L* constant	
	D3 use cords of the same material/density	
	D4 use CRO to determine *f* (or *T*)	
	D5 method to determine *T* from CRO, e.g. period = time base × length of one wave	
	D6 $f = 1 / T$	
	D7 repeat measurement of diameter along cord and average	
	D8 use of $A = \frac{\pi d^2}{4}$	
	D9 measure mass of the load on top-pan balance	
	D10 detail on determining frequency at the maximum amplitude, e.g. increase frequency until the amplitude starts to decrease, then decrease frequency	

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